Jiten Dhandha

PhD student - University of Cambridge

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Research interests

My research focuses on the study of the early Universe, spanning the **Dark Ages, Cosmic Dawn, and the Epoch of Reionization** (redshift $z \sim 6 - 50$). My expertise lies in the theoretical modelling of the **cosmic 21-cm signal** of neutral hydrogen, and of **stellar and galactic astrophysics**; this includes numerical simulations of present-day ($z \sim 0$) Population I star-formation in molecular clouds, as well as analytic and semi-numerical simulations of high-redshift ($z \gtrsim 6$) Population II and Population III stars. I have extensively used **machine learning and Bayesian inference techniques** to accelerate simulations and perform parameter inference, combining data from multi-wavelength facilities (e.g. from JWST and 21-cm experiments).

Employment

2026 - *present*	**Kavli Postdoctoral Fellow**, Kavli Institute for Physics and Mathematics of the Universe (Kavli IPMU), University of Tokyo.

Education

2022 - 2026	**PhD in Astronomy**, Institute of Astronomy, University of Cambridge. Funded by Boustany Astronomy Scholarship & Isaac Newton Studentship at Pembroke College. Supervised by Prof. Anastasia Fialkov and Dr. Eloy de Lera Acedo. **Thesis**: Constraining the Early Universe with 21-cm and Multi-Wavelength Inference in the JWST and SKA Era.
2018 - 2022	**MPhys. Physics with Astrophysics** (First Class), University of Manchester. Project involved simulating turbulent molecular clouds in ISM and studying filament and star formation. Performed with Zoe Faes and supervised by Dr. Rowan Smith.

Internships

Jul. 2024 - Aug. 2024	**Summer volunteer internship**, Boustany Foundation, Monaco. Partnering with Open Cultural Center, a humanitarian NGO focused on providing teaching, advice and extracurricular activities to asylum-seekers and refugees in the Nea Kavala camp in Greece.
Jun. 2021 - Aug. 2021	**Summer research project**, University of Manchester. Modelling the cosmological 21-cm signal in `Recfast++` and `CosmoTherm` to study their synergy with CMB spectral distortions. Supervised by Prof. Jens Chluba.
Jun. 2020 - Sep. 2020	**Summer research project**, University of Manchester. Testing and debugging `LOFAR-VLBI` calibration/imaging pipeline for gravitational lenses. Supervised by Dr. Neal Jackson.
Jul. 2019 - Sep. 2019	**Summer intern programme**, British Petroleum / University of Manchester. Simulating mitigation techniques for sulphate reducing bacteria responsible for fouling crude oil. Supervised by Dr. Thomas Waigh.

Grants and awards

April 2024	**DiRAC Resource Allocation Committee 16th Call**, awarded 4.15M CPUh (worth £41,500) on DiRAC's COSMA-8 supercomputer.
July 2022	**Tessella Prize for Software** (£125), for outstanding work implementing software in Mphys project.
March 2022	**Boustany Scholarship for Astronomy** ($\sim$ £100,530), for PhD at University of Cambridge.
April 2019	**BP Achievement Award** (£1000), for best essay on petrophysical logging tools.
December 2018	**Physics Success Scholarship** (£2000), for academic excellence in physics and maths.

Services

Conference, Workshop and Meeting organization

March 2026 - *present*	21-cm Meeting Series, co-organiser of bi-weekly meetings at KICC, University of Cambridge, focused on the cosmological 21-cm signal and related topics.
May 2026	Galaxies and the 21-cm Frontier: Unveiling the Epoch of Reionization (Kavli Science Focus), member of Organising Committee and session chair, KICC, University of Cambridge.
July 2025	Radio cosmology and science with the 21-cm signal (one day workshop), member of Organising Committee and session chair, KICC, University of Cambridge.
February 2024	Science with the 21-cm line (Kavli Science Focus), member of Organising Committee and session chair, KICC, University of Cambridge.

Other committees

Feb. 2025 - Mar. 2025	**International Womens Day Committee** member, Institute of Astronomy, UoC.
May 2023 - Oct. 2025	**EDI Inclusion and Fairness** subgroup member, Institute of Astronomy, UoC.
Jul. 2023 - Jul. 2024	**Graduate Parlour**, Ethnic Minorities officer, Pembroke College, UoC.
Oct. 2022 - Oct. 2025	**Postgraduate Forum** representative, Institute of Astronomy, UoC.
Oct. 2022 - Apr. 2023	**Pembroke Papers** committee memeber, Pembroke College, UoC.

Teaching responsibilities

Oct. 2025 - Sep. 2026	**Co-Supervision: Seowon Jung** (Part III and summer student) with Prof. Anastasia Fialkov and Prof. Alexander Belyaev. Work on probing the nature of dark matter with 21-cm cosmology.
Oct. 2025 - Jul. 2026	**Co-Supervision: Saughn Sekhon** (Part III student) with Prof. Anastasia Fialkov and Dr. Peter Sims. Work on machine learning accelerated Bayesian inference of early Universe star-formation.
Oct. 2024 - Jul. 2025	**Co-Supervision: Jacques Valkenberg** (MPhil. student) with Prof. Anastasia Fialkov and Prof. Sandro Tacchella. Work on studying the impact of metal enrichment and Population II stellar IMF on the 21-cm signal of neutral hydrogen.
Oct. 2024 - Jul. 2025	**Co-Supervision: Kyle Wong** (Part III student) with Prof. Anastasia Fialkov. Work on studying the impact of varying cosmology and matter power spectrum on the 21-cm signal of neutral hydrogen.
Oct. 2023 - Jul. 2024	**Co-Supervision: Jamie Incley** (Part III student) with Prof. Anastasia Fialkov. Work on comparison of Epoch of Reionization in simulation codes `21cmSPACE` and `C2-Ray`.
Feb. 2023 - Mar. 2023	**Demonstration of Part IA Scientific Computing** for 22 hours, University of Cambridge.

Talks

Invited talks

April 2026	*Constraining Galaxies in the Early Universe using 21-cm and Multi-Wavelength Synergies* (virtual), State of the Universe (SOTU) seminar, DTP, TIFR-Mumbai.
November 2025	*Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints*, Weekly seminar, Institut d'Astrophysique Spatiale, Université Paris-Saclay.
September 2025	*An overview of 21-cm cosmology and how we can constrain the discovery space of the 21-cm signal* (virtual), 8th Global 21-cm Workshop, California Institute of Technology.
October 2022	*Can accreting primordial black holes explain the excess radio background?* (virtual), Weekly webinar, PDAT Laboratory, K. N. Toosi University of Technology.

Conference and Workshop talks

November 2026	*Rotating Pop III stars and their impact on the high-z Universe*, Simulations GEARS, Kavli Institute for Physics and Mathematics of the Universe, University of Tokyo.
March 2026	*Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints*, Cosmology in the Alps 2026, Les Diablerets, Switzerland.
June 2025	*Exploiting synergies between JWST and cosmic 21-cm observations to uncover star formation in the early Universe*, SKAO General Science Meeting 2025, Görlitz, Germany.

October 2024	*Constraining star-formation efficiency in the early Universe using JWST and the cosmic 21-cm signal*, Introduction to KICC, Kavli Institute for Cosmology, University of Cambridge.
September 2024	*Constraining star-formation efficiency in the early Universe using JWST and the cosmic 21-cm signal*, 7th Global 21-cm Workshop, Raman Research Institute.
May 2024	*Synergies between 21-cm experiments and JWST observations*, Reionization in Relic Radiation (R3), Institut d'Astrophysique Spatiale, Université Paris-Saclay.
February 2024	*Synergies between 21-cm experiments and JWST observations*, One day workshop: Science with the 21-cm line, Kavli Institute for Cosmology, University of Cambridge.
September 2023	*FIlamEntary STructure Analysis (fiesta)*, AREPO-ISM workshop, University of Manchester.

Meeting talks

October 2026	*What do we know about the cosmic 21-cm signal so far, and how can we improve upon it?*, Astro seminar, Kavli Institute for Physics and Mathematics of the Universe, University of Tokyo.
April 2026	*Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints*, NenuFAR Cosmic Dawn plenary, Kavli Institute for Cosmology, University of Cambridge.
October 2025	*An overview of 21-cm cosmology and how we can constrain the discovery space of the 21-cm signal*, mini-workshop on 21-cm signal (headed by Prof. Naoki Yoshida), Department of Physics, University of Tokyo.
September 2025	*What do we know about the 21-cm signal so far?*, Gravitational Wave Astrophysics Group group meeting (headed by Prof. Michela Mapelli), Centre for Astronomy of Heidelberg University.
September 2023	*Bringing 21-cm simulations to the JWST era*, 2nd REACH Annual Meeting, University of Malta.

Outreach talks

October 2022	*Like beads on a string... Where do massive stars in our Universe come from? A brief look into studying our cosmos*, Pembroke Papers, Pembroke College, University of Cambridge.

Workshops attended

September 2025	IWR School on Machine Learning for Fundamental Physics, Faculty of Mathematics and Computer Science, Heidelberg University.

Software development

`eor_limits`	Author and maintainer of the new beta-release of the web-app: eorlimits.streamlit.app, which allows users to compare upper limits on the 21-cm power spectrum from different experiments.
`21cmSimulators`	Author and maintainer: Comprehensive public repository of popular 21-cm signal simulation codes.
`21cmExperiments`	Author and maintainer: Comprehensive public repository of past and ongoing 21-cm experiments.
`CFit`	Author and maintainer: Smart curve fitting tool using method of least squares in Python.
`fiesta`	Author and maintainer: Toolkit for analyzing filament networks and density field meshes.

In the media

August 2021	Most detailed-ever images of galaxies revealed using LOFAR. Press release for LOFAR observations from ASTRON.
August 2021	Astronomers develop novel way to 'see' first stars through fog of early Universe. Press release for LOFAR observations from BBC.

Extracurricular activities

Sep. 2021 - Jul. 2022	**Student Representative** representing astronomy/astrophysics, UoM.
Jul. 2020 - Jul. 2022	**Touch Rugby Society**, Inclusion officer and COVID-19 safety officer, UoM.
Sep. 2019 - Jun. 2020	**Peer-Assisted Study Session** leader, Peer Support Scheme,UoM.

Nov. 2016 - *present* | **English Wikipedia**, volunteer editor.

Publications

I have **3 first author** publications and **15 contributing author** publications. I have also contributed by peer-reviewing **1** paper.

First Author

December 2025 | **J. Dhandha**, A. Fialkov, T. Gessey-Jones, H. T. J. Bevins, S. Tacchella, S. Pochinda, E. de Lera Acedo, S. Singh, R. Barkana, *Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints*, Monthly Notices of the Royal Astronomical Society, 544, 1608

September 2025 | **J. Dhandha**, A. Fialkov, T. Gessey-Jones, H. T. J. Bevins, S. Tacchella, S. Pochinda, E. de Lera Acedo, S. Singh, R. Barkana, *Exploiting synergies between JWST and cosmic 21-cm observations to uncover star formation in the early Universe*, Monthly Notices of the Royal Astronomical Society, 542, 2292

April 2024 | **J. Dhandha**, Z. Faes, R. J. Smith, *Decaying turbulence in molecular clouds: how does it affect filament networks and star formation?*, Monthly Notices of the Royal Astronomical Society, 529, 4699

Contributing Author

June 2026 | S. A. K. Leeney et al., *Scalable Bayesian data curation for next-generation radio experiments*, arXiv e-prints, arXiv:2606.16525

April 2026 | J. Wasserman, E. Zackrisson, **J. Dhandha**, A. Fialkov, L. Noble, S. Majumdar, *Ultraviolet photon production rates of the first stars: Impact on the He II λ 1640 Å emission line from primordial star clusters and the 21-cm signal from cosmic dawn*, Monthly Notices of the Royal Astronomical Society, 547, stag386

April 2026 | S. Dasgupta, B. Liu, A. Fialkov, F. Deng, **J. Dhandha**, R. Barkana, *Impact of Stochastic Pop III X-ray Binaries on the Cosmological 21-cm Signal*, arXiv e-prints, arXiv:2604.11542

March 2026 | A. K. Dash et al., *Optimisation of calibration sources for global 21-cm experiments: the REACH case*, arXiv e-prints, arXiv:2604.00105

February 2026 | Z. Abdurashidova et al., *First Results from HERA Phase II*, The Astrophysical Journal, 998, 33

— 2026 | G. V. C. Allen et al., *Circuit Modeling for In Situ 21 cm Radiometer Calibration*, Journal of Astronomical Instrumentation, 15, 2650002-70

— 2026 | A. Chakraborty, T. R. Choudhury, K. K. Datta, P. Dayal, **J. Dhandha**, S. Gagnon-Hartman, S. K. Giri, A. Gorce, C. Heneka, A. Hutter, B. Maity, S. Majumdar, A. Mesinger, K. Moriwaki, C. S. Murmu, Y. Qin, S. Yoshiura, E. Zackrisson, *Synergies for the Epoch of Reionization and Cosmic Dawn*, Advancing Astrophysics with the SKA I, 137

October 2025 | S. Munshi, F. G. Mertens, J. K. Chege, L. V. E. Koopmans, A. R. Offringa, B. Semelin, R. Barkana, **J. Dhandha**, A. Fialkov, R. Mériot, S. Sikder, A. Bracco, S. A. Brackenhoff, E. Ceccotti, R. Ghara, S. Ghosh, I. Hothi, M. Mevius, P. Ocvirk, A. K. Shaw, S. Yatawatta, P. Zarka, *Improved upper limits on the 21-cm signal power spectrum at $z = 17.0$ and $z = 20.3$ from an optimal field observed with NenuFAR*, Monthly Notices of the Royal Astronomical Society, 542, 2785

August 2025 | B. Liu, D. Kessler, T. Gessey-Jones, **J. Dhandha**, A. Fialkov, Y. Sibony, G. Meynet, V. Bromm, R. Barkana, *Effects of chemically homogeneous evolution of the first stars on the 21-cm signal and reionization*, Monthly Notices of the Royal Astronomical Society, 541, 3113

March 2025 | O. S. D. O'Hara, Q. Gueuning, E. de Lera Acedo, F. Dulwich, J. Cumner, D. Anstey, A. Brown, A. Fialkov, **J. Dhandha**, A. Faulkner, Y. Liu, *Uncovering the effects of array mutual coupling in 21-cm experiments with the SKA-Low radio telescope*, Monthly Notices of the Royal Astronomical Society, 538, 31

February 2025 | S. Pochinda, **J. Dhandha**, A. Fialkov, E. de Lera Acedo, *Cosmological super-resolution of the 21-cm signal*, Accepted at NeurIPS 2024 (Machine Learning and the Physical Sciences Workshop), arXiv:2502.00852

September 2024 | O. S. D. O'Hara, F. Dulwich, E. de Lera Acedo, **J. Dhandha**, T. Gessey-Jones, D. Anstey, A. Fialkov, *Understanding spectral artefacts in SKA-Low 21-cm cosmology experiments: the impact of cable reflections*, Monthly Notices of the Royal Astronomical Society, 533, 2876

May 2024 | A. Fialkov, T. Gessey-Jones, **J. Dhandha**, *Cosmic mysteries and the hydrogen 21-cm line: bridging the gap with lunar observations*, Philosophical Transactions of the Royal Society of London Series A, 382, 20230068

December 2022 | S. K. Acharya, **J. Dhandha**, J. Chluba, *Can accreting primordial black holes explain the excess radio background?*, Monthly Notices of the Royal Astronomical Society, 517, 2454

February 2022	S. Badole, D. Venkattu, N. Jackson, S. Wallace, **J. Dhandha**, P. Hartley, C. Riddell-Rovira, A. Townsend, L. K. Morabito, J. P. McKean, *High-resolution imaging with the International LOFAR Telescope: Observations of the gravitational lenses MG 0751+2716 and CLASS B1600+434*, Astronomy and Astrophysics, 658, A7

White Papers / Reports / Book Chapters

June 2026	A. Mesinger et al., *Synergies for the Epoch of Reionization and Cosmic Dawn*, Advancing Astrophysics with the SKA II (AASKAII), 2026
November 2025	E. de Lera Acedo et al., *21 cm Cosmology from Space: A UK Path via CosmoCube*, UK Space Frontiers 2035
November 2025	J. Chluba et al., *Opening a New Window for Cosmology with CMB Spectral Distortions*, UK Space Frontiers 2035